Assignment 7: DynamoDB Change Alert with SNS

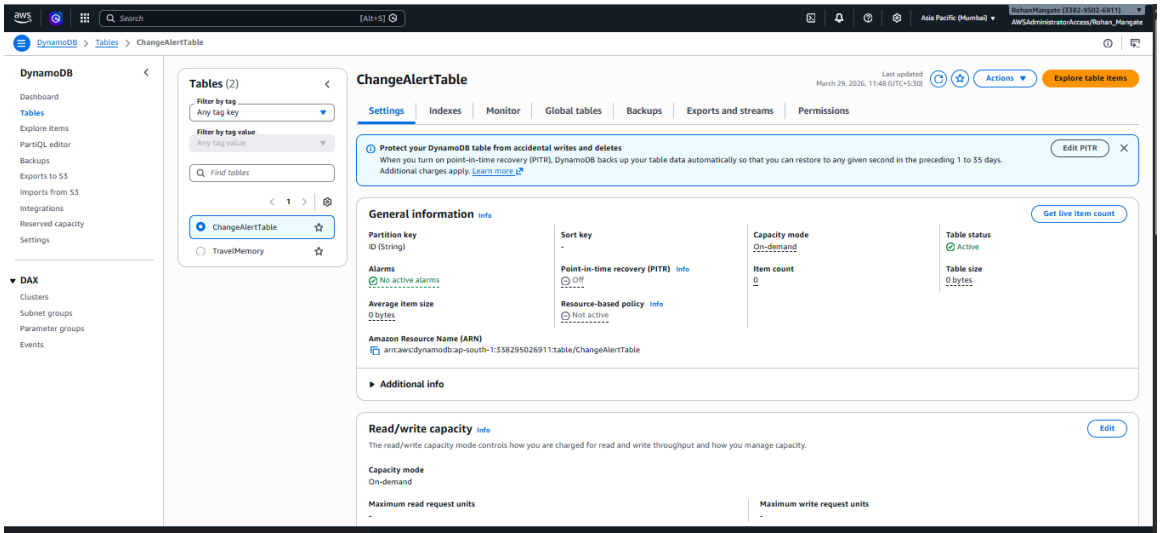
Objective:

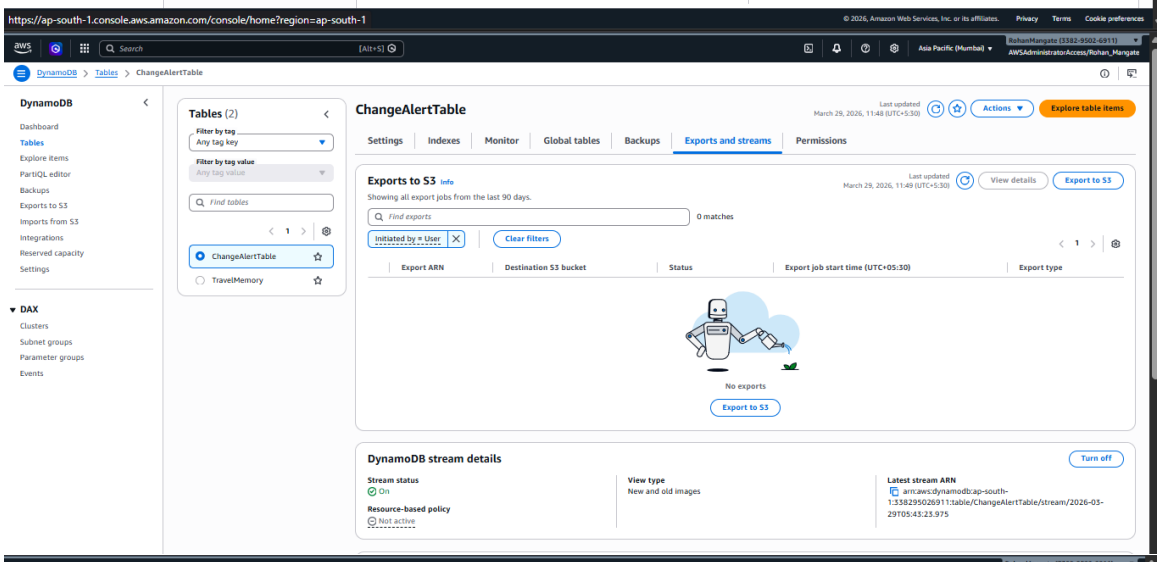
The objective of this assignment is to configure Amazon DynamoDB Streams with AWS Lambda to detect data changes (INSERT, MODIFY, DELETE) in a DynamoDB table and send real-time notifications using Amazon SNS email alerts.

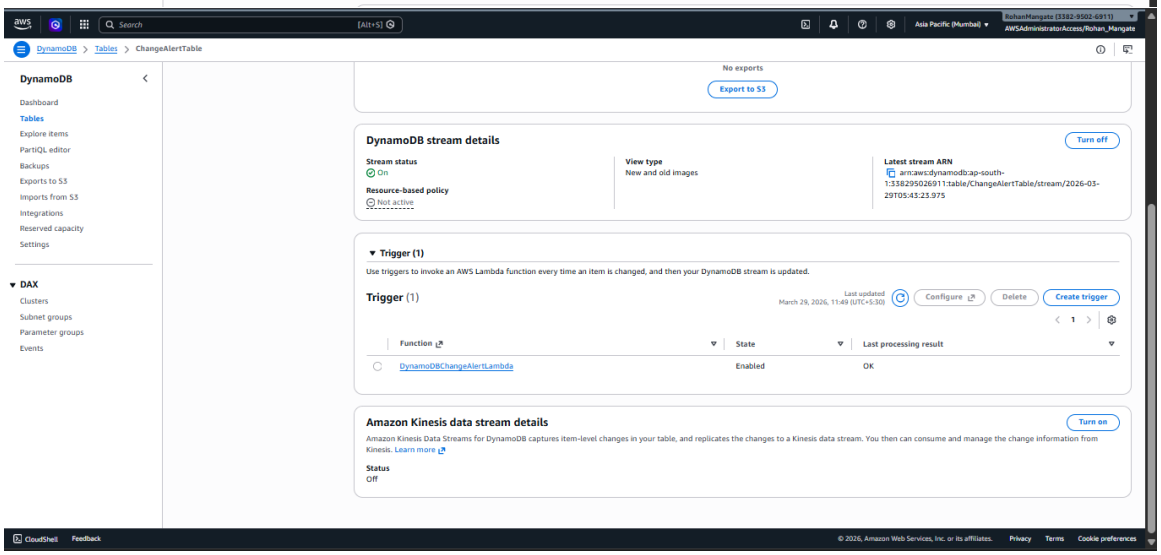
Step 1: DynamoDB Table Creation

A DynamoDB table named ChangeAlertTable was created with partition key id (String).
DynamoDB Streams were enabled with the view type set to New and Old Images.

Screenshot: DynamoDB table details and streams configuration.



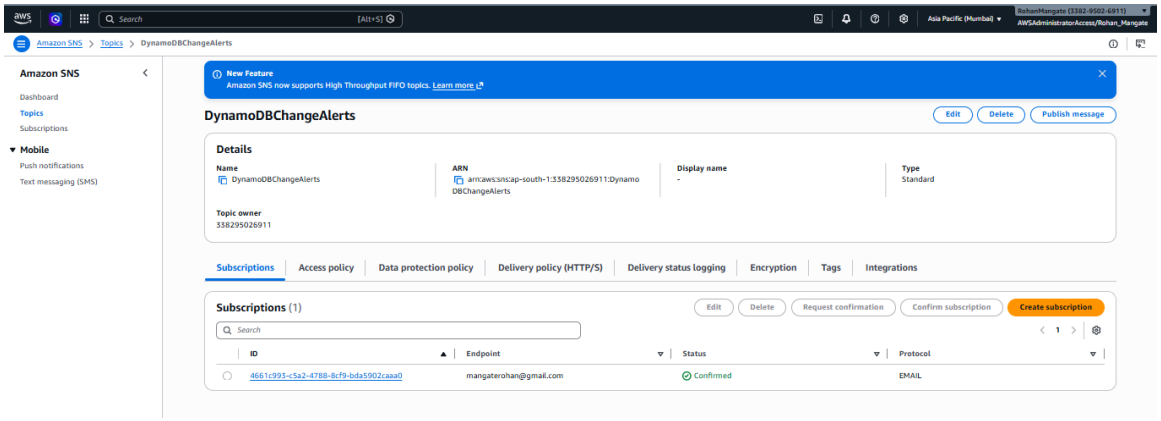




Step 2: SNS Topic and Subscription

An SNS topic named DynamoDBChangeAlerts was created.
An email subscription was added and successfully confirmed to receive notifications.

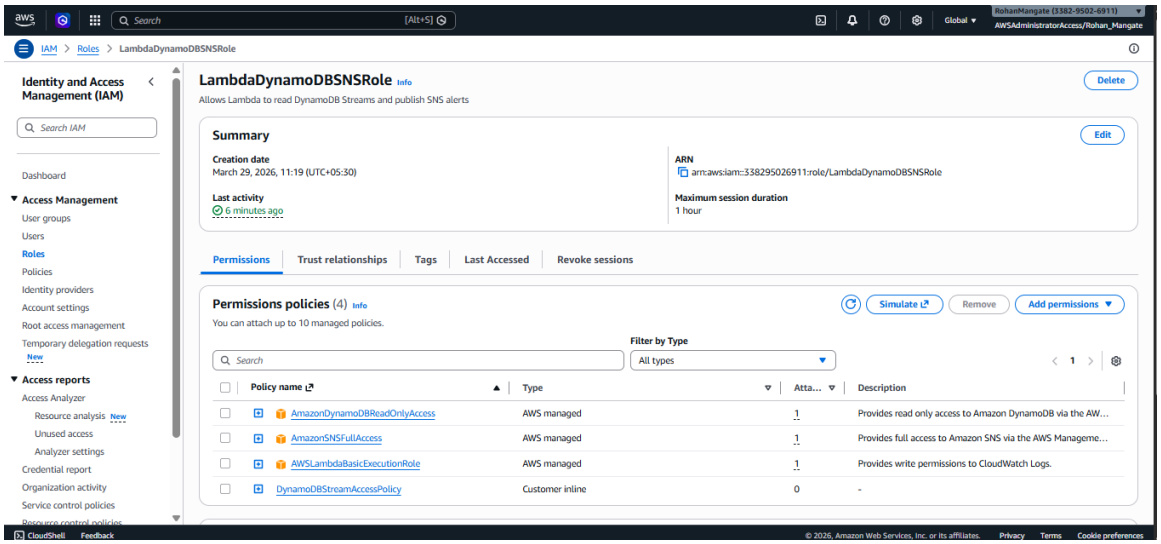
Screenshot: SNS topic details and confirmed subscription.



Step 3: IAM Role Creation

An IAM role named LambdaDynamoDBSNSRole was created for Lambda execution.
The role includes permissions for CloudWatch Logs, DynamoDB Streams access, and SNS publish actions.

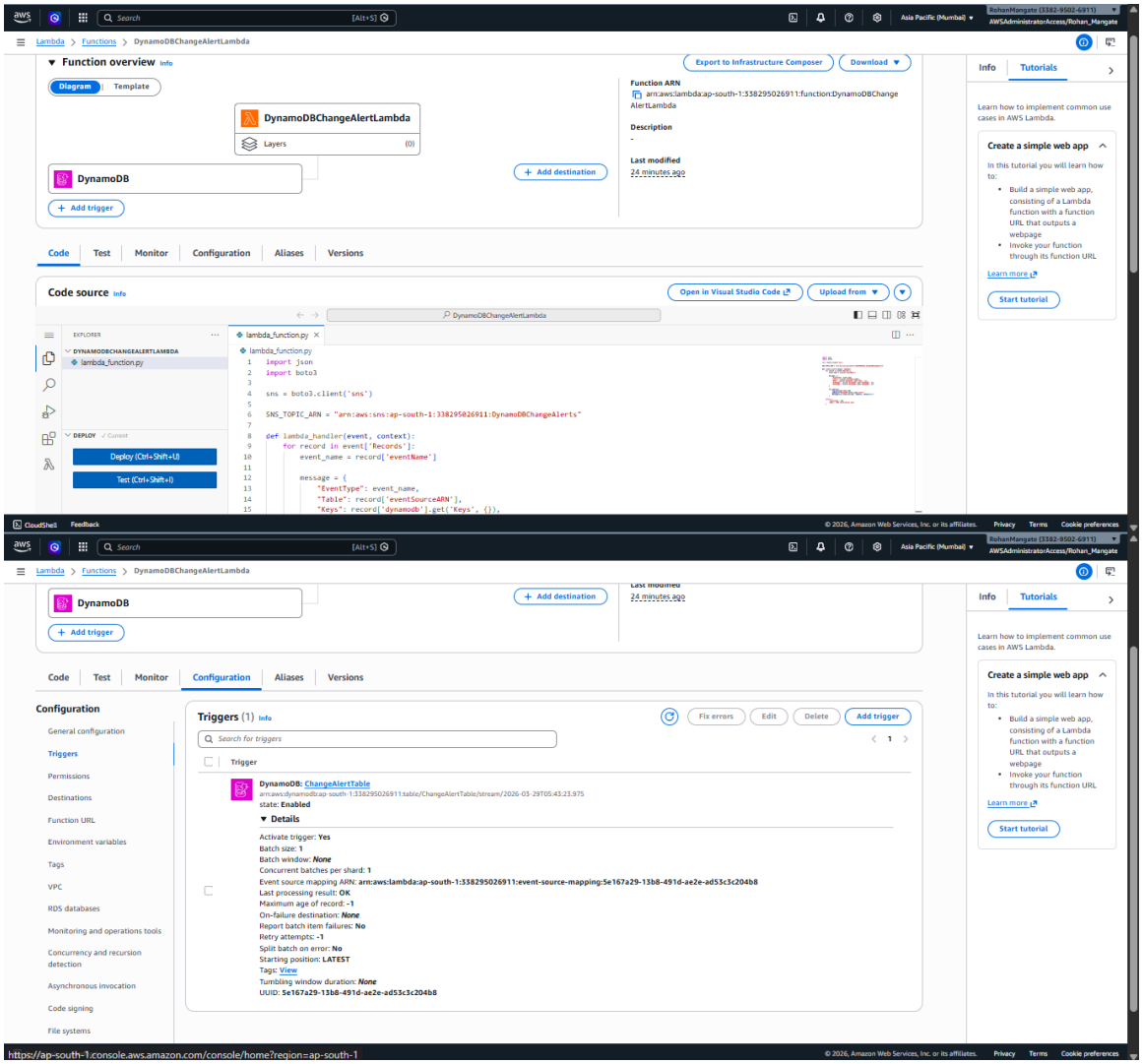
Screenshot: IAM role with attached policies.



Step 4: Lambda Function and Trigger Configuration

A Lambda function DynamoDBChangeAlertLambda was created using Python runtime.
The function was attached to the DynamoDB Stream trigger for ChangeAlertTable.

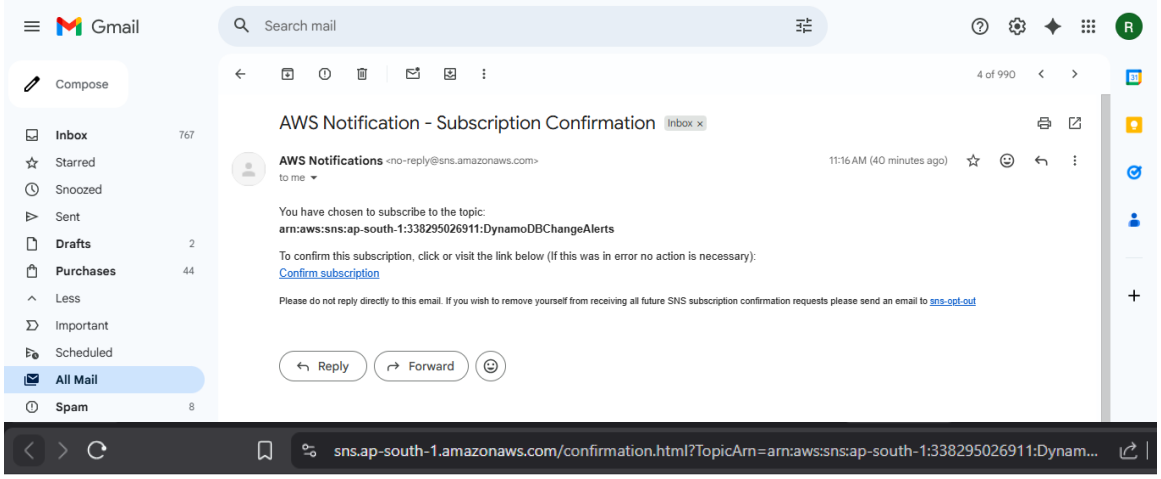
Screenshot: Lambda function details, trigger configuration, and function code.

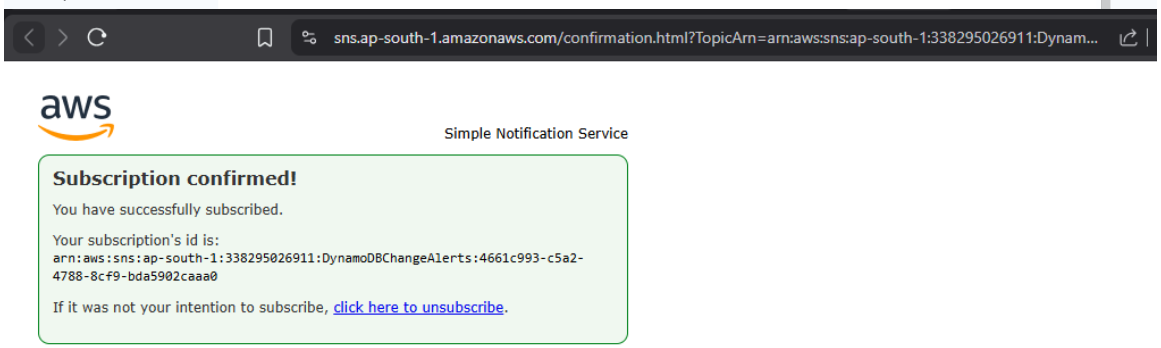


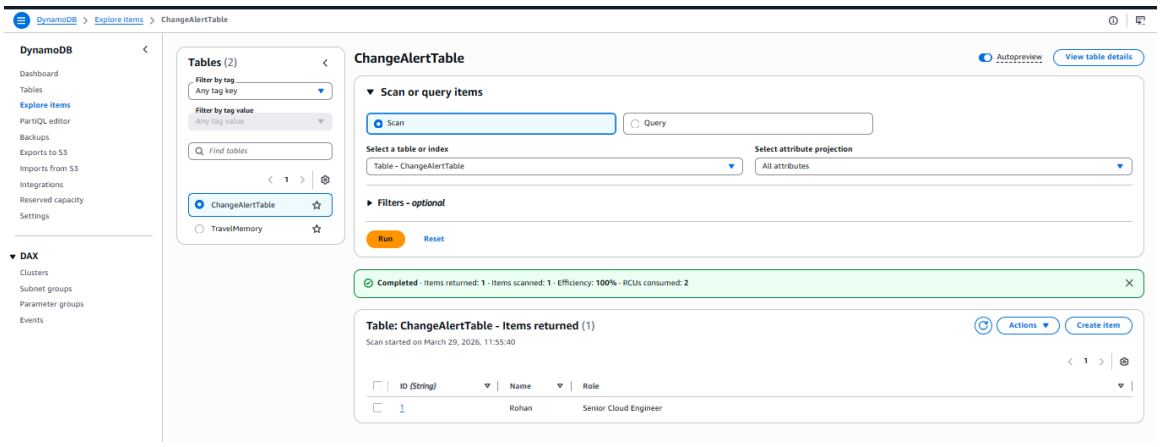
Step 5: Testing and Validation

Insert, Modify, and Delete operations were performed on the DynamoDB table.
Each operation successfully triggered the Lambda function and sent an SNS email alert.

Screenshots: DynamoDB operations and received email alerts.







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1–50 of 990

AWS Notifications	Inbox	DynamoDB MODIFY Alert - { "EventType": "MODIFY", "Table": "arn:aws:dynamodb:ap-south-1:...	11:34 AM
AWS Notifications 3	Inbox	DynamoDB INSERT Alert - { "EventType": "INSERT", "Table": "arn:aws:dynamodb:ap-south-1:33...	11:33 AM
AWS Notifications 2	Inbox	DynamoDB REMOVE Alert - { "EventType": "REMOVE", "Table": "arn:aws:dynamodb:ap-south-1...	11:33 AM
AWS Notifications	Inbox	AWS Notification - Subscription Confirmation - You have chosen to subscribe to the topic: arn:...	11:16 AM

DynamoDB REMOVE Alert

Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com>
to me

11:32 AM (24 minutes ago)


```
{
  "EventType": "REMOVE",
  "Table": "arn:aws:dynamodb:ap-south-1:338295026911:table/ChangeAlertTable/stream/2026-03-29T05:43:23.975",
  "Keys": {
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      "S": "1"
    }
  },
  "OldImage": {
    "ID": {
      "S": "1"
    }
  },
  "NewImage": {}
}
```


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https://sns.ap-south-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:ap-south-1:338295026911:DynamoDBChangeAlerts:4661c993-c5a2-4788-8cf9-bda5902caaa0&Endpoint=mangaterohan@gmail.com

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```
{
  "EventType": "REMOVE",
  "Table": "arn:aws:dynamodb:ap-south-1:338295026911:table/ChangeAlertTable/stream/2026-03-29T05:43:23.975",
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      "S": "2"
    }
  },
  "OldImage": {
    "Role": {
      "S": "Cloud Engineer"
    },
    "ID": {
      "S": "2"
    },
    "Name": {
      "S": "Rohan"
```


...

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